

WEEK 2 PROJECT REPORT

Advanced Data Visualization and Storytelling with Python

Student Performance Dataset

Project Domain	Education / Student Performance Analytics
Dataset Size	1,000 students × 8 variables
Visualization Tools	Python, Matplotlib, Seaborn
Focus	Visual narrative, comparisons, distributions and relationships
Submission	Week 2 Data Visualization and Storytelling Report

Prepared as a professional Week 2 internship submission

1. Introduction

This report presents an advanced visualization and data storytelling analysis of the Student Performance Dataset. The dataset contains academic scores together with demographic, family, lunch and test-preparation information. The objective of this week's work is not only to create charts, but also to transform numerical results into a clear visual narrative that can be understood by a non-technical audience.

The analysis focuses on six complementary visualizations: average scores by gender, score distribution, correlation among subjects, the effect of test preparation, mathematics scores by parental education level, and mathematics scores by lunch type. Together, these charts move the story from simple comparison to distribution, relationship and contextual interpretation.

2. Dataset Description

The dataset contains 1,000 student records and 8 columns: gender, race/ethnicity, parental level of education, lunch type, test preparation course, math score, reading score and writing score. The three score variables are numerical and are measured on a 0–100 scale. The categorical variables allow performance to be compared across different student groups.

Variable Group	Examples	Purpose in Analysis
Demographic	Gender, race/ethnicity	Compare student groups
Family / support	Parental education	Explore contextual differences
Resources	Lunch type	Compare performance distributions
Preparation	Test preparation course	Assess performance differences
Academic	Math, reading, writing	Measure learning outcomes

3. Methodology

The analysis was performed in Google Colab using Python. Pandas was used for data handling and aggregation, while Matplotlib and Seaborn were used to create the visualizations. Grouped means were used where category-level comparison was appropriate. Histograms were used to understand score distributions, a correlation heatmap was used to reveal relationships between academic subjects, and a box plot was used to compare score spread and outliers between lunch groups.

The visualization choices were made to support a coherent story: first understand who performs better on average, then examine the overall distribution, identify relationships among subjects, study the influence of preparation, and finally examine contextual factors associated with math performance.

3.1 Average Student Scores by Gender

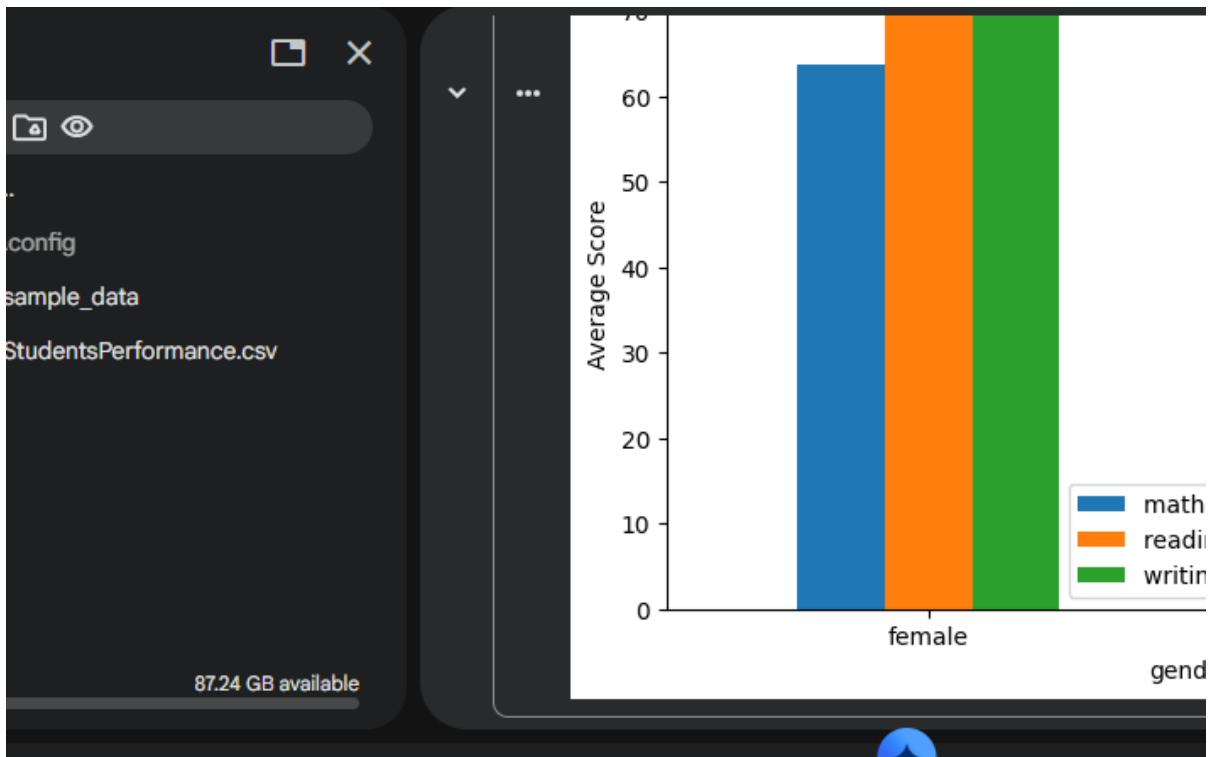


Figure: 1 Average Student Scores by Gender

Why this visualization was chosen

A grouped bar chart was selected because it makes category-level differences easy to compare across the three subjects.

Key insight

Female students show higher average reading and writing scores, while male students show a higher average mathematics score. The differences are visible but not extreme, suggesting that subject-specific patterns are more informative than a single overall ranking.

3.2 Distribution of Student Scores

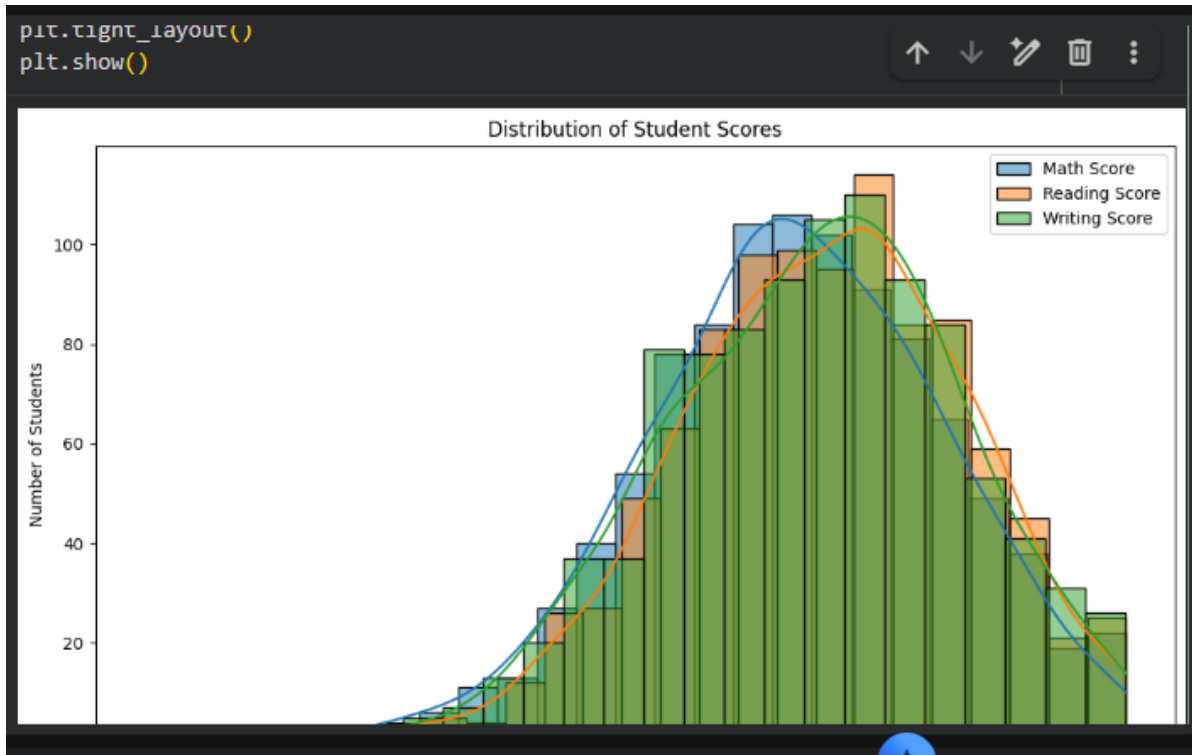


Figure: 2 Distribution of Student Scores

Why this visualization was chosen

A combined histogram with smooth density curves was used to show how math, reading and writing scores are distributed across the 1,000 students.

Key insight

All three subjects show broadly concentrated distributions around the middle-to-upper score range. Reading and writing distributions overlap strongly, while mathematics has a similar but slightly different shape. This indicates that most students cluster around moderate to good performance rather than being evenly spread across the full 0–100 range.

3.3 Correlation Between Student Scores



Figure: 3 Correlation Between Student Scores

Why this visualization was chosen

A heatmap was chosen because correlation coefficients can be compared quickly through both color intensity and displayed values.

Key insight

Reading and writing have the strongest relationship at about 0.95. Mathematics is also strongly related to reading (about 0.82) and writing (about 0.80). The result suggests that students who perform well in one academic subject tend to perform well in the others.

3.4 Impact of Test Preparation on Student Performance

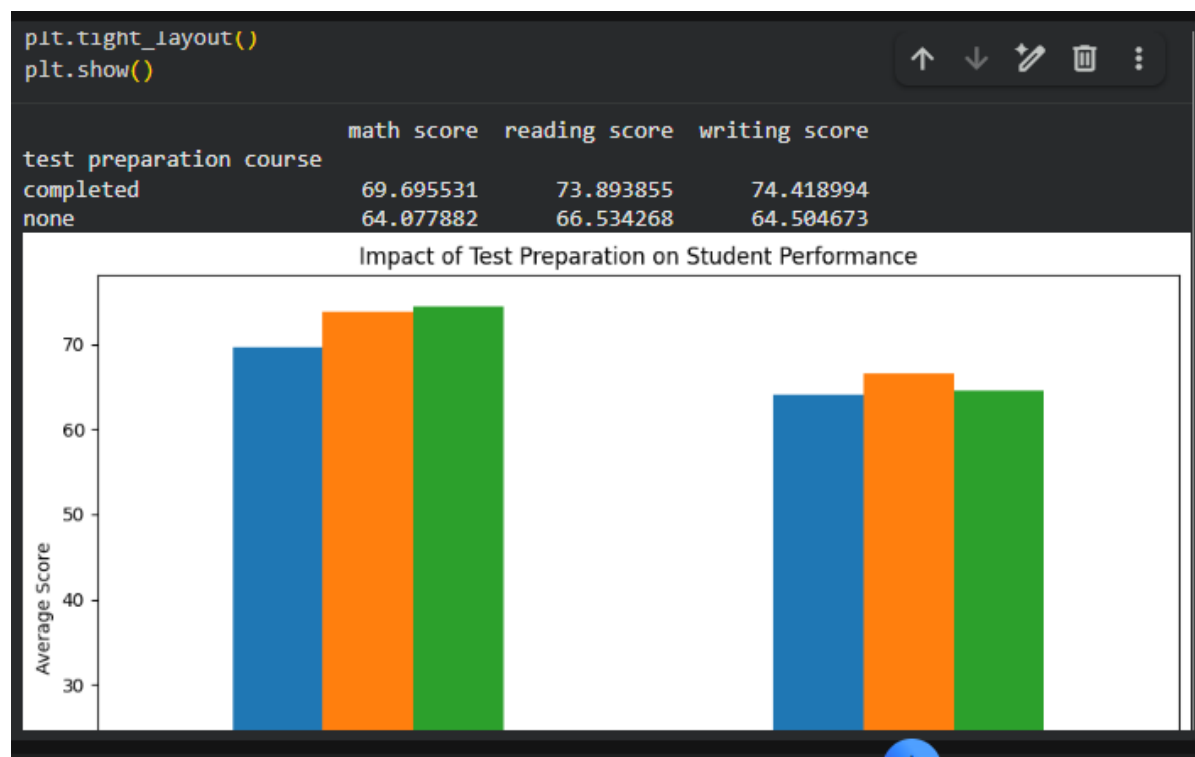


Figure: 4 Impact of Test Preparation on Student Performance

Why this visualization was chosen

A grouped bar chart was used to compare average mathematics, reading and writing scores for students who completed the preparation course and those who did not.

Key insight

Students who completed test preparation have higher average scores in all three subjects. The improvement is especially visible in reading and writing, while mathematics also shows a positive difference. This makes test preparation a useful factor to highlight in the data story, while recognizing that the chart shows association rather than proof of causation.

3.5 Math Score by Parental Level of Education

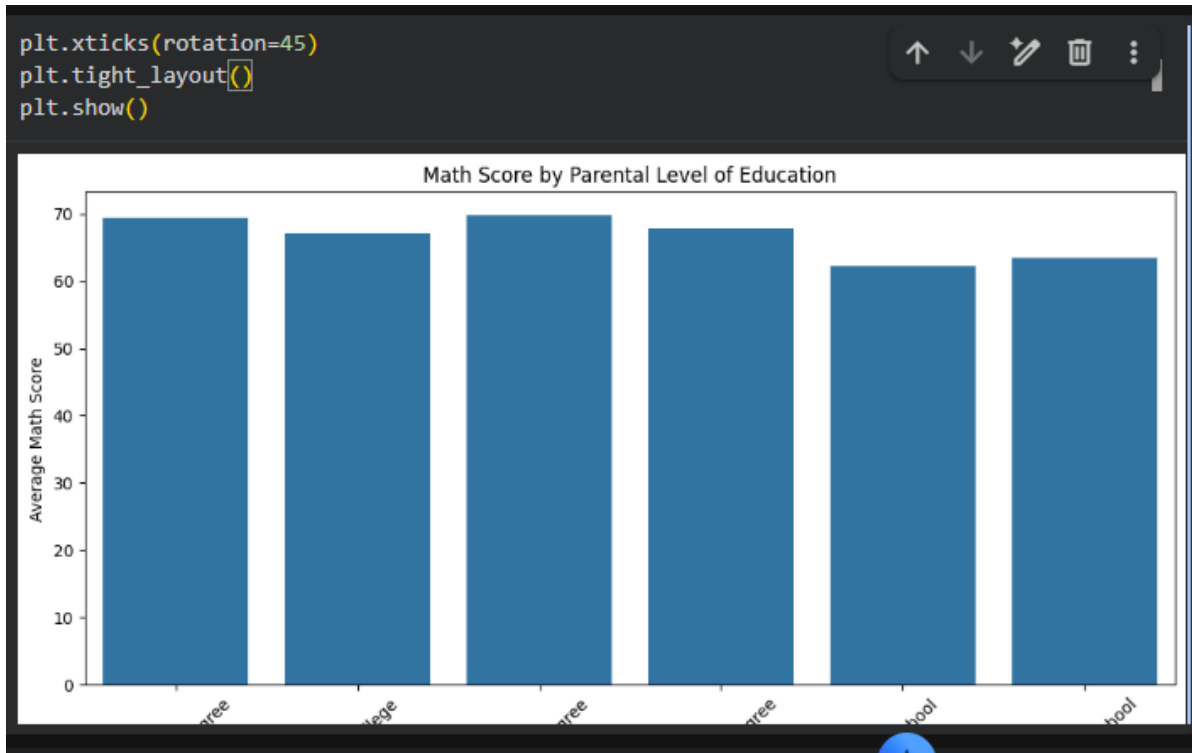


Figure: 5 Math Score by Parental Level of Education

Why this visualization was chosen

A category comparison bar chart was used to examine whether average mathematics performance varies across parental education levels.

Key insight

The averages vary across education categories. Groups associated with higher levels of parental education tend to show relatively stronger mathematics averages, while some lower-education categories have lower averages. The differences are moderate, so parental education should be interpreted as a contextual factor rather than a direct cause.

3.6 Math Score Distribution by Lunch Type

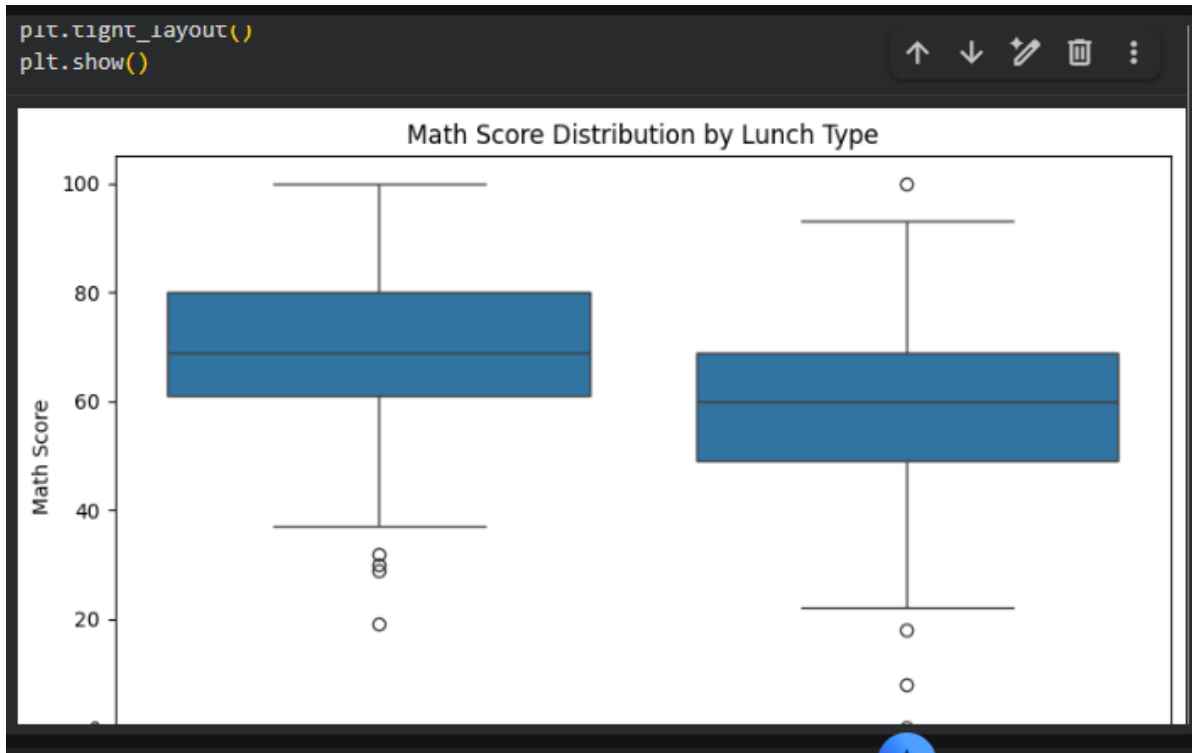


Figure: 6 Math Score Distribution by Lunch Type

Why this visualization was chosen

A box plot was selected because it shows the median, middle 50% of scores, overall spread and potential outliers at the same time.

Key insight

Students receiving standard lunch have a higher central math score than students receiving free/reduced lunch. The free/reduced group also shows a wider lower tail and several low-score outliers. This indicates a meaningful difference in the distribution, not just in the average.

10. Overall Data Story

The visualizations form a connected story about student performance. The first chart establishes that performance differs by subject and gender, but no single group dominates every subject. The distribution chart then shows that most scores are concentrated in a relatively consistent range. The correlation heatmap provides a stronger structural insight: performance across subjects is closely connected, particularly reading and writing.

The next two visualizations introduce possible contextual influences. Test preparation is associated with higher average scores across all three subjects, while parental education shows moderate differences in mathematics performance. Finally, the lunch box plot reveals a clear separation in math-score distributions between standard and free/reduced lunch groups. The combined narrative is therefore that academic outcomes are interconnected and are also associated with preparation and student-context variables.

11. Business and Scientific Implications

- Educational institutions can use performance dashboards to identify groups that may benefit from additional academic support.
- Test-preparation programs can be evaluated using before/after or controlled studies in future work; the current analysis only identifies an association.
- Strong correlations between subjects suggest that improvement in one learning area may be useful to investigate alongside progress in related subjects.
- Lunch-group differences indicate that socioeconomic or resource-related factors may be worth investigating when designing student-support programs.
- Parental education patterns can help researchers explore broader educational context, but conclusions should avoid treating background characteristics as deterministic causes.

12. Key Findings at a Glance

Finding	Evidence from Visualization
Reading and writing are highly connected.	Correlation ≈ 0.95
Math is strongly related to reading and writing.	Correlation ≈ 0.82 and 0.80
Test preparation is associated with higher scores.	Higher averages in all three subjects
Standard lunch group has a higher math-score center.	Box plot shows higher median
Performance differs across parental-education groups.	Math averages vary by category
Gender differences depend on the subject.	Female higher in reading/writing; male higher in math

13. Conclusion

This project demonstrates how Python-based visualization can turn a structured student dataset into an understandable data story. Six different visualization techniques were used to answer different analytical questions rather than repeating the same chart type. The analysis highlights strong relationships among academic subjects and identifies meaningful performance differences associated with preparation and contextual variables.

The main lesson from the visualization story is that student performance should be viewed from multiple dimensions. Average scores alone do not explain the full picture; distributions, correlations and group-level context provide additional evidence. For future work, the analysis could be extended with interactive Plotly dashboards, statistical significance testing, regression models and longitudinal data to distinguish association from causation.

14. Tools and Technologies

- Python
- Google Colab
- Pandas
- Matplotlib
- Seaborn
- Student Performance Dataset

15. Final Submission Checklist

✓ Introduction and dataset description included

- ✓ Methodology and visualization rationale included
- ✓ Six visualizations included with captions and interpretations
- ✓ Data storytelling section included
- ✓ Business/scientific implications included
- ✓ Conclusion and future scope included
- ✓ Report formatted as a professional Word document